

DCTPR: Recovering Most Transductive Gains with Three Support-Anchored Prototype Updates

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Abstract

Frozen visual foundation models provide strong transferable representations, but a single labeled image per class remains an unreliable prototype in high-way fine-grained recognition. Transductive solvers can use the unlabeled query set to correct this bias, although their iterative optimization can be costly. We present Dual-Capacity Transductive Prototype Refinement (DCTPR), a training-free task adapter that concatenates normalized frozen DINOv2-B/14 and DINOv2-L/14 features, estimates class-balanced query centers, and performs three fixed prototype updates. We evaluate the frozen procedure on three 200-way CUB episodes and three 120-way Stanford Dogs episodes, with ten support rotations per episode. DCTPR improves dual-capacity nearest support classification by 6.68 and 6.27 percentage points on CUB and Dogs, respectively. Its accuracy remains within 1.35 points of TIM-ADM on CUB and 0.61 points of MAP-RAW on Dogs, while its transductive head is 3.8 and 49 times faster than those strongest matched solvers. Controlled class-imbalance tests show that this efficiency depends materially on a known balanced query prior. Relative to nearest support, DCTPR recovers 83.3% and 91.1% of the accuracy gain delivered by the strongest matched solver on CUB and Dogs. The results position DCTPR as a reproducible accuracy–efficiency operating point for balanced, high-way, one-shot fine-grained recognition rather than as an accuracy state-of-the-art method.

Keywords: few-shot learning; transductive inference; fine-grained recognition; frozen visual foundation models; prototype refinement

1 Introduction

Fine-grained recognition distinguishes visually similar categories while remaining robust to pose, viewpoint, background, and within-class appearance variation. These properties make the problem particularly sensitive to label scarcity. When only one labeled image represents each class, the support image is both the supervision signal and the initial class prototype. Atypical pose or background content can therefore shift the prototype away from the query distribution even when the underlying feature extractor is strong. Prior fine-grained few-shot methods therefore devote explicit machinery to local attention, fea-

ture reconstruction, or cross-image correspondence [1–3].

Large self-supervised visual encoders such as DINOv2 provide transferable representations without task-specific backbone training [4]. Their strength changes the practical few-shot question. In a moderate-shot regime, the representation and a simple classifier may already be difficult to improve, as established by transfer-based and nearest-centroid studies [5–7]. In a high-way one-shot regime, however, the dominant error can instead arise from estimating a class center from one observation. This distinction motivates task-level adaptation that corrects prototypes without fitting neural parameters.

Transductive few-shot methods jointly classify an unlabeled query set and can reduce single-support bias. LaplacianShot propagates information over a query graph [8]; TIM optimizes a mutual-information objective [9]; and PT-MAP combines feature transformation, optimal transport, and prototype updates [10]. These methods establish strong accuracy baselines, but their graph or iterative optimization also creates a practical latency cost. Moreover, many transductive benchmarks assume that every class has the same number of queries, an assumption that can strongly affect inference [11]. The unresolved engineering question is therefore not only which solver is most accurate, but how much of the transductive gain can be recovered by a short, transparent, and training-free update under an explicit query-prior scope.

This question calls for a different evaluation target from an accuracy-only leaderboard. A useful short adapter should produce a substantial improvement over the inductive reference, remain close to the strongest matched solver, and reduce the task-level work required for every new episode. These three requirements must be evaluated together. Reporting only the gain over nearest support would hide the remaining accuracy gap, whereas reporting only the best accuracy would hide the cost of reaching it. We therefore treat accuracy, runtime, cross-episode stability, and prior sensitivity as parts of one operating-point claim.

We address this question with Dual-Capacity Transductive Prototype Refinement (DCTPR). The method combines frozen DINOv2-B/14 and DINOv2-L/14 class-token features, imposes the known balanced query marginal through a soft Sinkhorn assignment, and alternates assignment with a support-anchored query-center update for exactly three refinement steps. The procedure does not read query labels and learns no task-specific parameters. It is evaluated under a frozen development–confirmation pro-

tocol on CUB and then transferred without retuning to Stanford Dogs.

Our contributions are threefold. First, we define a compact dual-capacity task adapter for balanced high-way one-shot recognition and specify its assumptions, normalization, assignment, and update rules completely. Second, we evaluate the method across two fine-grained datasets, six train-only episodes, and 60 support rotations against matched implementations of established transductive solvers. Third, we report both computational cost and controlled query-imbalance stress tests, establishing where the method is competitive and where its uniform-prior assumption fails.

2 Related Work

2.1 Frozen representations for few-shot recognition

Transfer-based few-shot learning often separates representation learning from task inference. A strong encoder can make simple nearest-center or linear classifiers competitive, which places a higher burden on any adaptation module to show that its gain is not caused by a weak feature baseline [5–7]. DINOv2 learns general-purpose visual features through self-supervised pretraining [4]. We therefore freeze both DINOv2 encoders in all experiments and apply every compared transductive solver to the same extracted features. This design isolates task inference from backbone optimization.

Fine-grained datasets add a complementary challenge: class identity can depend on small appearance differences while nuisance variation dominates the image. CUB [12] and Stanford Dogs [13] are widely used to study such distinctions, including in attention- and reconstruction-based few-shot recognition [1–3]. Rather than constructing a conventional 5-way task, we evaluate all available categories in each episode. This high-way setting magnifies competition among similar categories and makes support-prototype bias visible even with foundation-model features.

2.2 Transductive inference and prototype correction

Transductive methods use the query set jointly rather than predicting each query independently. LaplacianShot optimizes unary support distances together with a graph smoothness term [8]. TIM alternates classifier updates under supervised support and query information objectives [9]. PT-MAP applies a feature transform and Sinkhorn-based assignment before iteratively moving class prototypes [10]. Related work also shows that careful feature preprocessing and simple classifier ensembles can remain competitive without episodic meta-training [14,15]. More recent work further combines graph and prototype refinement [16] or learns conditional transport for unknown query proportions [17].

DCTPR does not claim the individual novelty of feature concatenation, Sinkhorn normalization, or prototype refinement. Its contribution is the scoped combination of complementary frozen capacities with a deliberately short update schedule, together with a matched high-way accuracy–efficiency evaluation. The method is consequently compared against the established solver families on the same feature tensors instead of against a weaker classifier alone.

2.3 Query priors in transductive evaluation

Balanced query sets provide useful controlled evaluation, but they also reveal the class marginal to the inference procedure. Veilleux et al. showed that methods encoding a uniform marginal can degrade under Dirichlet-distributed query proportions and proposed an alpha-entropy objective to reduce this bias [11]. Conditional transport and adaptive-manifold methods subsequently targeted imbalanced transduction directly [17,18]. We treat balance as an explicit assumption rather than a hidden property. Our primary experiments test the balanced high-way setting, while separate mild and severe imbalance regimes measure how far the conclusion extends.

3 Method

3.1 Task formulation

An episode contains C classes and one labeled support image per class, $\mathcal{S} = \{(x_c^s, c)\}_{c=1}^C$. The unlabeled query set is $\mathcal{Q} = \{x_i^q\}_{i=1}^N$. In the primary protocol, each class occurs exactly $q = N/C$ times in $\mathcal{Q}$, but query labels are never observed by the method. The goal is to predict all query labels jointly using frozen image features and the known balanced column mass q .

Figure 1 summarizes DCTPR. Two frozen encoders produce complementary image representations. The fused support features initialize the class prototypes. Each refinement step alternates a balanced soft assignment with a support-anchored query-center update. The final assignment provides the query predictions.

3.2 Design principles

DCTPR is designed around four constraints. First, all task information must enter through the support and unlabeled query features; the adapter must not update either encoder or learn task-specific neural parameters. Second, the support example must remain an explicit anchor because the initial soft assignments can contain errors. Third, the query-set constraint must match an assumption that can be stated and tested, rather than silently exploiting a balanced sampler. Fourth, task-level computation must be bounded by a small fixed number of matrix operations.

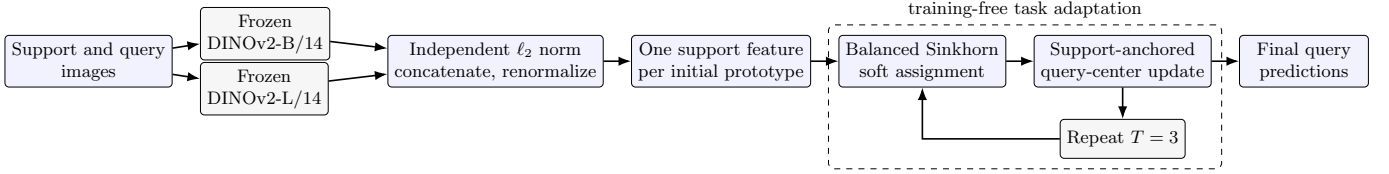


Figure 1: Overview of DCTPR. Both encoders remain frozen. The method uses the balanced query marginal but never accesses query labels. Only the assignment and support-anchored prototype update are repeated.

These constraints determine the pipeline in Figure 1. Independent normalization prevents the larger feature family from contributing only through scale. Balanced assignment uses the known query marginal to regularize an otherwise uncertain 200- or 120-way matching problem. Support-anchored updates incorporate the resulting query centers without recursively replacing the labeled evidence. Finally, fixing the schedule to three updates makes latency independent of a convergence tolerance. The component study tests the first three choices, while the matched runtime comparison tests whether the short schedule produces a useful operating point.

3.3 Dual-capacity representation

Let $f_B(x) \in \mathbb{R}^{d_B}$ and $f_L(x) \in \mathbb{R}^{d_L}$ denote the class-token outputs of frozen DINOv2-B/14 and DINOv2-L/14. Each capacity is normalized before concatenation so that one feature family cannot dominate only because of its raw norm:

$$\bar{f}_k(x) = \frac{f_k(x)}{\|f_k(x)\|_2}, \quad k \in \{B, L\}. \quad (1)$$

The fused representation is

$$z(x) = \frac{[\bar{f}_B(x); \bar{f}_L(x)]}{\|[\bar{f}_B(x); \bar{f}_L(x)]\|_2}. \quad (2)$$

The initial prototype is the only labeled representation for its class, $p_c^{(0)} = z(x_c^s)$. The ablation in Section 5.1 tests whether the fused representation contributes beyond either capacity alone.

3.4 Balanced soft assignment

At refinement step t , cosine logits compare every query with every prototype:

$$A_{ic}^{(t)} = \frac{z(x_i^q)^\top p_c^{(t)}}{\tau}, \quad \tau = 0.05. \quad (3)$$

We exponentiate the stabilized logits and apply Sinkhorn row and column scaling [19]. The resulting matrix $P^{(t)} \in \mathbb{R}_{+}^{N \times C}$ satisfies

$$P^{(t)} \mathbf{1}_C = \mathbf{1}_N, \quad (P^{(t)})^\top \mathbf{1}_N = q \mathbf{1}_C. \quad (4)$$

The row constraint produces a categorical distribution for each query, while the column constraint encodes the known balanced query count. We use 100 normalization iterations. Equation (4) is a task assumption, not a learned estimate; Section 5.4 tests its failure under unknown non-uniform counts.

3.5 Support-anchored prototype update

The soft query center for class c is

$$\mu_c^{(t)} = \frac{\sum_{i=1}^N P_{ic}^{(t)} z(x_i^q)}{\sum_{i=1}^N P_{ic}^{(t)}}. \quad (5)$$

We anchor every update to the original labeled support rather than recursively averaging the preceding prototype:

$$p_c^{(t+1)} = \text{norm}_2 \left((1 - \lambda) p_c^{(0)} + \lambda \mu_c^{(t)} \right), \quad \lambda = 0.5. \quad (6)$$

The assignment and update are repeated for $T = 3$ steps, after which a final balanced assignment yields the predicted class. This support anchor limits prototype drift, while the short fixed schedule bounds task-level latency. No gradient update, query label, or dataset-specific parameter is used after the CUB development episode.

For feature dimension $d = d_B + d_L$, one assignment/update step is dominated by the $N \times C$ similarity matrix and query-center aggregation, both $O(NCd)$. The Sinkhorn normalization acts on the smaller assignment matrix. In our implementation, the complete three-step transductive head takes about 33–34 ms per high-way support rotation on an RTX 3080 Ti, excluding the common frozen feature extraction stage.

4 Experimental Protocol

4.1 Datasets and episode construction

CUB-200-2011 contains 200 bird species [12]. Each CUB episode uses ten official-training images per class, producing a 200-way task with one support and nine queries per class. Three episodes are evaluated. Episode 1 is the only development episode; Episodes 2 and 3 are confirmation episodes.

Stanford Dogs contains 120 dog breeds [13]. We select three pairwise-disjoint groups of ten images per class from the official training list, for 3,600 unique images in total. Each group defines a 120-way one-shot episode with nine queries per class. All method and baseline constants are transferred from CUB without using Dogs labels for selection.

Figure 2 illustrates the task: related classes can differ in subtle plumage or shape while pose and background vary substantially.

Within each episode, the ten images of every class take turns as support. This produces ten paired rotations

a CUB-200-2011 official training**b Stanford Dogs official training**

Figure 2: **Official-training examples.** **a**, CUB-200-2011 warblers. **b**, Visually similar Stanford Dogs terriers. This context-only plate is not quantitative evidence. Images were split-verified and center-cropped without color or contrast adjustment; no official-test image is shown.

and ensures that every image is used as the sole support once. We report the mean balanced accuracy across rotations. Because rotations reuse episode images, their standard deviations are descriptive and are not treated as independent-sample confidence intervals. No CUB or Dogs official-test image was decoded or encoded during this study.

4.2 Features and implementation

Images are directly resized to 392×392 . We extract the class token from the official frozen DINOv2 ViT-B/14 and ViT-L/14 models. The same saved feature tensors are supplied to all inference methods. DCTPR uses $\tau = 0.05$, 100 Sinkhorn normalizations, $T = 3$ prototype updates, and $\lambda = 0.5$. These constants are frozen before CUB confirmation and before all Dogs experiments.

Runtime is measured only for the task-level transductive head on one NVIDIA GeForce RTX 3080 Ti; common feature extraction and disk loading are excluded. The reported time is the mean wall-clock duration per support rotation, with CUDA synchronization around the measured call.

4.3 Matched baselines

The inductive reference, BL nearest support (BL-NCC), predicts from cosine similarity in the fused feature space. BL-Sinkhorn adds balanced soft assignment without prototype refinement. Single-capacity refinement applies the same update to either B or L features.

We additionally implement TIM-ADM [9], raw MAP and signed-power PT-MAP [10], and LaplacianShot [8]. Their published update rules are applied to the common DINO features. TIM-ADM uses temperature 15, loss weights (0.1, 1.0, 0.1), 150 alternating updates, and update factor 1. MAP uses transport regularization 10, prototype update factor 0.2, and 20 updates. Since DINO class-token values are signed, the PT-MAP power-transform arm uses an explicitly labeled signed square-root adaptation. LaplacianShot uses directed non-self neighbors, 20 bound updates, and a published lambda grid selected only on CUB Episode 1. These are matched feature-level reimplementations, not executions of the original end-to-end

backbone pipelines.

4.4 Metrics and comparison logic

Balanced accuracy is the primary metric because every primary episode has the same number of queries per class and the stress test deliberately changes those counts. For the balanced episodes, balanced accuracy equals ordinary query accuracy. For the imbalanced regimes, we compute recall per class and average over classes so that frequent classes cannot dominate the result.

Runtime measures only the task-level transductive head. This isolates the computational difference among solvers because all methods receive the same saved DINOv2 features. It does not measure end-to-end image latency and should not be interpreted as such. BL-NCC has zero additional task-head optimization and is therefore reported as an inductive reference rather than placed at an artificial positive time on logarithmic plots.

To summarize how much of the strongest solver’s improvement is retained, we also report a descriptive gain-recovery ratio

$$R_{\text{gain}} = \frac{A_{\text{DCTPR}} - A_{\text{BL}}}{A_{\text{strong}} - A_{\text{BL}}}, \quad (7)$$

where A_{strong} is the accuracy of the strongest matched transductive solver on that dataset. This ratio complements, rather than replaces, the absolute accuracy and runtime values.

5 Results

5.1 Component analysis

Table 1 separates capacity fusion, balanced assignment, and prototype refinement. BL-NCC averages 68.17% across CUB. Balanced assignment without refinement raises this to 73.50%, showing that query-set geometry provides most of the initial gain. Refinement on B or L alone reaches 73.40% and 73.20%, respectively. Combining both capacities with assignment and support-anchored updates reaches 74.85%, 1.35 points above BL-Sinkhorn and 1.45–1.65 points above the single-capacity refinements.

Table 1: CUB component analysis. Values are balanced accuracy (%) averaged over ten rotations. E1 is development; E2–E3 are frozen confirmation.

Method	E1	E2	E3	Mean
BL nearest support	68.42	68.12	67.96	68.17
B-only refinement	73.84	73.22	73.13	73.40
L-only refinement	73.46	73.22	72.92	73.20
BL balanced assignment	73.79	73.44	73.26	73.50
DCTPR	75.14	74.71	74.69	74.85

The frozen confirmation is important because Episode 1 determined the fixed method constants. On Episodes 2 and 3, DCTPR exceeds the strongest component comparator by 1.27 and 1.43 points, respectively, while improving over BL-NCC by 6.59 and 6.73 points. Thus the full configuration does not depend on the development episode alone.

We also screened an agreement-weighted consensus extension on Episode 1. It reached 75.18%, only 0.04 points above its paired refinement reference, and was non-inferior in 6 of 10 rotations. It failed the frozen development gate and was not retained. This negative control favors the simpler fixed DCTPR update over an unconfirmed adaptive weighting mechanism.

5.2 Accuracy–efficiency comparison

Table 2 compares established transductive solvers using the same features and episodes. On CUB, TIM-ADM is the most accurate method at 76.19%. DCTPR reaches 74.85%, a deficit of 1.35 points, but takes 33.3 ms rather than 126.3 ms per rotation, a 3.8-fold reduction. DCTPR also exceeds both MAP arms and the LaplacianShot variants on CUB.

On Dogs, MAP-RAW is strongest at 70.97%, while DCTPR reaches 70.36%. The 0.61-point difference is accompanied by a substantial runtime difference: 34.1 ms for DCTPR and 1,678.6 ms for MAP-RAW, or approximately 49-fold. DCTPR is also 0.36 points above TIM-ADM on Dogs while remaining about four times faster. These comparisons support an accuracy–efficiency claim, but not an accuracy state-of-the-art claim.

Figure 3 makes the operating-point comparison explicit. On CUB, DCTPR lies between the inexpensive balanced assignment and the more accurate TIM-ADM solver. It recovers $(74.85 - 68.17)/(76.19 - 68.17) = 83.3\%$ of TIM-ADM’s gain over BL-NCC while using 26.4% of its task-head runtime. On Dogs, it recovers $(70.36 - 64.09)/(70.97 - 64.09) = 91.1\%$ of MAP-RAW’s gain while using 2.0% of its runtime. The result is therefore not merely that DCTPR is faster than the most accurate comparator. It retains most of that comparator’s improvement over the same inductive reference.

5.3 Cross-episode and cross-dataset stability

The gain over nearest support is consistent across all six episodes (Table 3). CUB gains range from 6.59 to 6.73 points. Dogs gains range from 5.94 to 6.58 points even though no Dogs label informed method or baseline selection. All 30 Dogs support rotations improve over BL-NCC. The similar gain magnitude across a bird dataset and a dog-breed dataset suggests that the method is correcting a recurring one-shot prototype problem rather than only exploiting one CUB episode construction.

The rotation-level view in Figure 4a strengthens this descriptive result. Every one of the 60 paired rotations improves over BL-NCC, despite different support images and disjoint episode image groups. The dispersion is wider within episodes than the variation among episode means: individual gains range from approximately four to nine points, whereas the six episode means remain between 5.94 and 6.73 points. Because rotations reuse images, these points are evidence of paired stability rather than independent replicates for a significance test.

5.4 Query-prior stress test

The primary protocol supplies the exact class-balanced column mass used by Equation (4). We test the consequence of violating this assumption on CUB. In the mild regime, half the classes retain three queries and half retain nine. In the severe regime, class counts range from one to nine. Counts are permuted across classes with fixed seeds, and macro balanced accuracy prevents majority classes from dominating the metric.

Table 4 shows that uniform-prior DCTPR falls to 68.34% and 68.86%, close to nearest support and below its balanced-query result. Replacing the uniform marginal with true class counts, solely as an oracle diagnostic, recovers 73.35% and 72.87%. The 4.01–5.01 point mean gaps indicate that incorrect class mass, rather than complete loss of query geometry, explains a large fraction of the degradation. Oracle counts are never used as a deployable method input.

Figure 4b places the stress result next to the balanced operating point. The uniform and oracle DCTPR curves coincide when the query set is balanced because the assumed and true class marginals are identical. They separate immediately when the class counts become non-uniform. This pattern localizes the failure to the marginal constraint: query features still contain recoverable information, but the uniform column mass allocates that information to the wrong class totals.

6 Discussion

The central result is not maximum accuracy, but that short support-anchored refinement recovers a large and stable fraction of the gap between nearest support and stronger optimization. Component analysis attributes

Table 2: Matched transductive comparison. Accuracy is the mean balanced accuracy across three episodes and 30 rotations per dataset. Runtime is mean milliseconds per rotation for the transductive head only. Bold marks the best accuracy; underlining marks DCTPR.

Method	CUB, 200-way		Stanford Dogs, 120-way	
	Accuracy (%)	Runtime (ms)	Accuracy (%)	Runtime (ms)
BL nearest support	68.17	0.0	64.09	0.0
BL balanced assignment	73.50	10.9	66.88	12.0
LaplacianShot	68.30	2.2	64.43	3.9
MAP-RAW	74.30	1,612.9	70.97	1,678.6
Signed PT-MAP adaptation	74.37	1,612.5	70.94	1,666.0
TIM-ADM	76.19	126.3	70.01	139.2
<u>DCTPR</u>	<u>74.85</u>	<u>33.3</u>	<u>70.36</u>	<u>34.1</u>

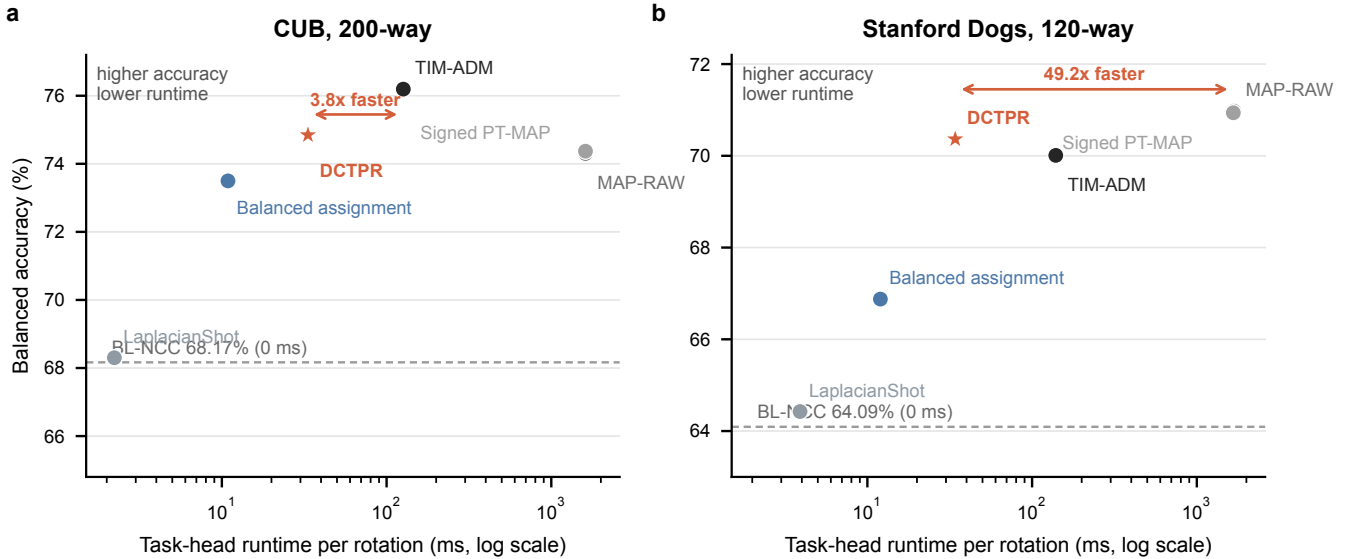


Figure 3: **Accuracy-efficiency operating points on two high-way datasets.** **a**, CUB-200-2011, evaluated as 200-way one-shot classification. **b**, Stanford Dogs, evaluated as 120-way one-shot classification. Each point reports mean balanced accuracy and mean task-head runtime across three episodes and 30 support rotations. All methods use the same saved DINOv2 features. The dashed horizontal line is the zero-optimization BL-NCC reference and is not assigned an artificial value on the logarithmic runtime axis. DCTPR is 3.8 times faster than TIM-ADM on CUB and 49.2 times faster than MAP-RAW on Dogs.

Table 3: Per-episode stability of the gain over fused nearest support.

Dataset/episode	BL-NCC	DCTPR	Gain (pp)
CUB E1	68.42	75.14	+6.72
CUB E2	68.12	74.71	+6.59
CUB E3	67.96	74.69	+6.73
Dogs E1	63.78	69.72	+5.94
Dogs E2	64.46	71.05	+6.58
Dogs E3	64.04	70.32	+6.29

this to balanced assignment, which corrects much of the single-support bias, and dual-capacity refinement, which adds 1.35 points. Both single-capacity variants are weaker than the fusion, although this does not establish a causal

Table 4: CUB macro balanced accuracy (%) under deterministic query imbalance, averaged over three episodes. Oracle counts are diagnostic only.

Method	Mild 3/9	Severe 1-9
BL nearest support	68.33	68.19
TIM-ADM	70.31	68.94
DCTPR, uniform prior	68.34	68.86
DCTPR, oracle counts	73.35	72.87

semantic mechanism.

The gain-recovery view clarifies the magnitude of this result. On the two datasets, the three-step adapter retains 83.3–91.1% of the improvement that the strongest

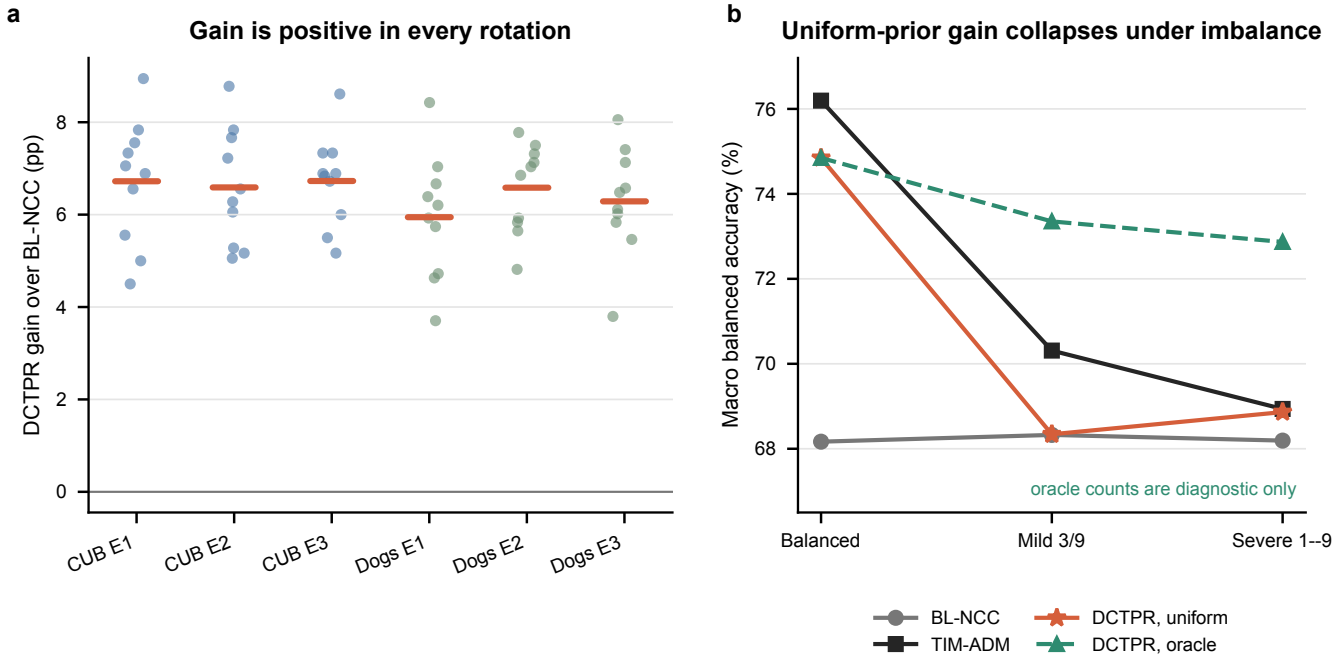


Figure 4: **Stable one-shot gains and their query-prior boundary.** **a**, Paired DCTPR gain over BL-NCC for each support rotation. Dots are the ten rotations per episode and orange bars are episode means; rotation dispersion is descriptive because rotations reuse images. All 60 differences are positive. **b**, Mean CUB macro balanced accuracy across the balanced protocol and two deterministic imbalance regimes. Uniform-prior DCTPR loses most of its balanced-task advantage, whereas supplying the true class counts recovers much of the loss. Oracle counts are a diagnostic and are never used as a deployable method input.

matched solver obtains over nearest support. This is more informative than comparing DCTPR only with BL-NCC, because it accounts for the accuracy still available from heavier optimization. It is also more conservative than calling the method Pareto-optimal in a universal sense: the frontier depends on the evaluated solver set, hardware, feature tensors, and episode construction.

The runtime comparison defines the useful operating regime. TIM-ADM adds 1.35 points on CUB at 3.8 times the task-head runtime; MAP-RAW adds 0.61 points on Dogs at about 49 times the runtime. Maximum-accuracy offline use may prefer those solvers, whereas repeated support selection or interactive recognition can amortize feature extraction and benefit from the three-step head. This is an engineering tradeoff, not universal dominance, and it does not imply that head latency dominates end-to-end image encoding.

The dual-capacity result should also be interpreted narrowly. Fusion improves over both single-capacity refinements on all three CUB episodes, and the frozen configuration transfers to Dogs without retuning. This is consistent with useful geometric complementarity between the B and L representations. Nevertheless, concatenation increases stored feature dimensionality, and the experiments do not identify which semantic attributes are complementary. Accordingly, dual capacity is supported as an empirical design choice within the adapter, not as a demon-

strated representation mechanism.

The balanced query prior is the main scope limitation. Under both imbalance regimes, forcing uniform class mass removes most of the balanced-task advantage. This agrees with broader evidence that class-balanced transductive benchmarks can favor methods that encode a uniform marginal [11]. Future extensions should estimate class proportions without query labels and should be evaluated on public Dirichlet-imbalance protocols against specialized methods such as conditional transport and adaptive manifolds [17, 18]. Such an extension is separate from the balanced-prior claim established here.

Three additional boundaries affect interpretation. First, the episodes use official training partitions to preserve an untouched official test; the reported numbers are therefore controlled train-only protocol results rather than standard leaderboard results. Second, established baselines are matched reimplementations of their update rules on common DINO features, not full executions of their original pretrained pipelines. Third, support rotations share images, so rotation variability is descriptive. These choices strengthen paired control and reproducibility but limit direct comparison with published results under different backbones, splits, and task samplers.

Two evaluations are therefore needed before extending the claim: a public imbalanced protocol with estimated class proportions and matched backbones, and end-

to-end latency measurements across hardware and batch sizes. They would test whether the short-update advantage survives unknown marginals and image encoding, respectively.

7 Conclusion

We introduced DCTPR, a training-free dual-capacity adapter for balanced, high-way, one-shot fine-grained recognition. Across CUB and Stanford Dogs, the method improves fused nearest-support classification by 6.68 and 6.27 percentage points, respectively, while remaining within 0.61–1.35 points of the strongest matched transductive solvers at substantially lower task-level runtime. Equivalently, it retains 83.3–91.1% of the strongest solver’s gain over nearest support while using 2.0–26.4% of that solver’s runtime. Component and frozen cross-dataset results support the use of short support-anchored query-center refinement as a practical accuracy–efficiency choice. The conclusion is explicitly bounded by a known balanced query prior; controlled imbalance tests show that removing this assumption requires a different inference design.

Reproducibility Statement

The project archive contains immutable experiment protocols, feature metadata, per-rotation predictions, summary JSON files, source hashes, and independent recomputation reports. Both quantitative figures are reconstructed from those archives, and the image plate uses six split-verified, hashed training samples. Figure source-data CSV files and the Python generation script accompany the draft. All method constants are stated in Sections 3 and 4, and all official-test access counters remain zero.

References

- [1] Yaohui Zhu, Chenlong Liu, and Shuqiang Jiang. Multi-attention meta learning for few-shot fine-grained image recognition. In *Proceedings of the Twenty-Ninth International Joint Conference on Artificial Intelligence*, pages 1090–1096, 2020.
- [2] Shu-Lin Xu, Faen Zhang, Xiu-Shen Wei, and Jianhua Wang. Dual attention networks for few-shot fine-grained recognition. *Proceedings of the AAAI Conference on Artificial Intelligence*, 36(3):2911–2919, 2022.
- [3] Jijie Wu, Dongliang Chang, Aneeshan Sain, Xiaoxu Li, Zhanyu Ma, Jie Cao, Jun Guo, and Yi-Zhe Song. Bi-directional feature reconstruction network for fine-grained few-shot image classification. *Proceedings of the AAAI Conference on Artificial Intelligence*, 37(3):2821–2829, 2023.
- [4] Maxime Oquab, Timothee Darcet, Theo Moutakanni, Huy Vo, Marc Szafraniec, Vasil Khalidov, Pierre Fernandez, Daniel HAZIZA, Francisco Massa, Alaaeldin El-Nouby, Mahmoud Assran, Nicolas Ballas, Wojciech Galuba, Russell Howes, Po-Yao Huang, Shang-Wen Li, Ishan Misra, Michael Rabbat, Vasu Sharma, Gabriel Synnaeve, Hu Xu, Herve Jegou, Julien Mairal, Patrick Labatut, Armand Joulin, and Piotr Bojanowski. DINOv2: Learning robust visual features without supervision. *Transactions on Machine Learning Research*, 2024.
- [5] Wei-Yu Chen, Yen-Cheng Liu, Zsolt Kira, Yu-Chiang Frank Wang, and Jia-Bin Huang. A closer look at few-shot classification. In *International Conference on Learning Representations*, 2019.
- [6] Yonglong Tian, Yue Wang, Dilip Krishnan, Joshua B. Tenenbaum, and Phillip Isola. Rethinking few-shot image classification: A good embedding is all you need? In *Computer Vision – ECCV 2020*, Lecture Notes in Computer Science, pages 266–282, 2020.
- [7] Yan Wang, Wei-Lun Chao, Kilian Q. Weinberger, and Laurens van der Maaten. SimpleShot: Revisiting nearest-neighbor classification for few-shot learning. In *International Conference on Learning Representations*, 2020.
- [8] Imtiaz Ziko, Jose Dolz, Eric Granger, and Ismail Ben Ayed. Laplacian regularized few-shot learning. In *Proceedings of the 37th International Conference on Machine Learning*, volume 119 of *Proceedings of Machine Learning Research*, pages 11660–11670, 2020.
- [9] Malik Boudiaf, Imtiaz Ziko, Jerome Rony, Jose Dolz, Pablo Piantanida, and Ismail Ben Ayed. Transductive information maximization for few-shot learning. In *Advances in Neural Information Processing Systems*, volume 33, 2020.
- [10] Yuqing Hu, Vincent Gripon, and Stephane Pateux. Leveraging the feature distribution in transfer-based few-shot learning. In *Computer Vision – ECCV 2020 Workshops*, Lecture Notes in Computer Science, pages 487–499, 2021.
- [11] Olivier Veilleux, Malik Boudiaf, Pablo Piantanida, and Ismail Ben Ayed. Realistic evaluation of transductive few-shot learning. *arXiv preprint arXiv:2204.11181*, 2022.
- [12] Catherine Wah, Steve Branson, Peter Welinder, Pietro Perona, and Serge Belongie. The caltech-ucsd birds-200-2011 dataset. Technical Report CNS-TR-2011-001, California Institute of Technology, 2011.
- [13] Aditya Khosla, Nityananda Jayadevaprakash, Bangpeng Yao, and Li Fei-Fei. Novel dataset for fine-grained image categorization: Stanford dogs. In *CVPR Workshop on Fine-Grained Visual Categorization*, 2011.
- [14] Yuqing Hu, Stephane Pateux, and Vincent Gripon. Squeezing backbone feature distributions to the max for efficient few-shot learning. *Algorithms*, 15(5):147, 2022.
- [15] Yassir Bendou, Yuqing Hu, Raphael Lafargue, Giulia Lioi, Bastien Passet, Stephane Pateux, and Vincent Gripon. EASY—ensemble augmented-shot-y-shaped learning: State-of-the-art few-shot classification with simple components. *Journal of Imaging*, 8(7):179, 2022.
- [16] Hao Zhu and Piotr Koniusz. Transductive few-shot learning with prototype-based label propagation by iterative graph refinement. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 23996–24006, 2023.
- [17] Long Tian, Jingyi Feng, Wenchao Chen, Xiaoqiang Chai, Liming Wang, Xiyang Liu, and Bo Chen. Prototypes-oriented transductive few-shot learning with conditional transport. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 16271–16280, 2023.
- [18] Michalis Lazarou, Yannis Avrithis, and Tania Stathaki. Adaptive manifold for imbalanced transductive few-shot learning. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, pages 2286–2295, 2024.
- [19] Marco Cuturi. Sinkhorn distances: Lightspeed computation of optimal transport. In *Advances in Neural Information Processing Systems*, volume 26, 2013.